

dr. kornpob bhirombhakdi

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Summary

AI Security Researcher, Educator, and Consultant — dual PhDs (Astrophysics + Mechanism Design & Game Theory), NASA postdoc, red-teaming lead, author of the *Spellbook of Jailbreak and Agent Vulnerabilities*. High-stakes AI safety, agent vulnerabilities, adversarial ML.

- ☐ dual PhDs + NASA postdoc
- ☐ red-teaming / adversarial research lead
- ☐♂ author, *Spellbook of Jailbreak and Agent Vulnerabilities*
- ☐ Winner, *The Judgement Day 2026 Phase-1 Scenario Submission*

Quick Links

- ☐ bkornpob@gmail.com
- ☐ bkornpob.github.io
- ☐ kbhirombhakdi.weebly.com
- ☐ ORCID: 0000-0003-0136-1281

recent bits

Spellbook of Jailbreak and Agent Vulnerabilities (Zenodo:21665118 + Page + Spotify + Youtube)

The Judgement Day 2026 Phase-1 Scenario Submission (GitHub + Spotify + Youtube)

Goodboy Recon (Github + Medium)

ZADDY breadcrumbs 2026 collection (Zenodo:21791515 + Spotify + Substack)

Education

☐ Ph.D. Physics

Ohio University (2014-2019)

Light Curve Powering Mechanism of Superluminous Supernovae

☐ Ph.D. Economics

Chulalongkorn University (2009-2013)

Game Theory & Experimental Studies on Reciprocity

□ **M.S. Physics**

Ohio University (2014-2015)

GPA: 3.951

□ **M.S. Health Economics**

Chulalongkorn University (2007-2009)

Technical Efficiency of University Hospitals

□ **B.S. Medical Science**

Siriraj, Mahidol University (2003-2007)

Second Class Honours

□ **Physics Pre-Degree**

Ramkhamhaeng University (2013-2014)

GPA: 4.0

Certifications & Training

Certified in Cybersecurity

ISC2 — ID: 3153050

Jan 2026 - Dec 2028

BASI Research Contributor

Oct 2025 - present

Cybersecurity Trainings

THNCA+NCSA Thailand 2025

Courses: CompTIA Cloud+, Computer Hacking Forensic Investigator, Basic Penetration Testing

Cybersecurity Professional Course

NCSA Thailand — ID: 5241057415KP

2025

Basic Cybersecurity

NCSA Thailand — ID: SPG-ZNEXJM1UDTRGX

2025

Deep Learning Specialization

DeepLearning.AI (Coursera) — ID: GYIBDXJ9UMCP

2025

Data Science and Machine Learning Specialization

Johns Hopkins University (Coursera) 2015 - 2017

Courses: - Developing Data Products - Practical Machine Learning - Regression Models - Exploratory Data Analysis - Getting and Cleaning Data - Reproducible Research - R Programming - Statistical Inference - The Data Scientist's Toolbox

Employment

On-Demand Consultant — Business Analytics & AI Solutions

DR.KB (Self-Employed), Bangkok, Thailand

Sep 2025 - Present

- **Strategic Advisory:** Consult C-suite and product leads on AI integration, risk mitigation, and enterprise analytics roadmaps.
- **Technical Implementation:** Design, build, and deploy custom machine learning models, LLM agents, and automated data pipelines.
- **Cross-Functional Leadership:** Bridge technical engineering and executive decision-making to drive ROI and project execution.

Lead Prompt Security Researcher

InjectPrompt.com (Contract), Remote

Jan 2026 - May 2026

- **Red-Teaming Architecture:** Led red-team research on AI agent safety and jailbreak tactics to strengthen model defense interfaces and system resilience.
- **Workflow & Knowledge Systems:** Designed custom RAG knowledge bases, skills, and orchestration workflows for sandboxed jailbreak testing.
- **Research-to-Product:** Authored and released *InjectPrompt Companion 3.0* and the *Jailbreak Perturbation Framework (JPF)* to deploy automated AI engines for dynamic jailbreak generation and guardrail robustness testing.

Data Scientist

Sertis Co., Ltd., Bangkok, Thailand

Aug 2024 - Aug 2025

- **Computer Vision:** Developed deep learning architectures for fine-grained object identification and multi-object tracking in video streams.
- **Time-Series Forecasting:** Built scalable multi-branch forecasting models for retail and wholesale supply chains, optimizing inventory allocation and demand prediction across multiple retail locations.

- **ML Asset Maintenance & Governance:** Maintained and optimized enterprise-grade reusable code repositories and ML assets to ensure continuous reliability, scalability, and seamless deployment across internal teams.
- **Provisional Technical Advisory:** Delivered expert consulting during client engagements, resolving complex technical bottlenecks on demand and pitching high-value extension solutions to drive business expansion.
- **Technical Client Relations & Strategy:** Served as the direct bridge between technical stakeholders, enterprise clients, and internal business development teams, leading cross-functional alignment and project roadmapping.

Postdoctoral Researcher

Space Telescope Science Institute (AURA/NASA), Baltimore, MD, USA

Aug 2019 – Feb 2024

Research/Teaching Assistant

Physics and Astronomy Department, Ohio University, Athens, OH, USA

Aug 2014 – May 2019

Senior Consultant

LEARN Corporation Co., Ltd., Bangkok, Thailand

Jan 2012 – Jan 2014

Market Research Consultant

Freelance, Bangkok, Thailand

Jan 2010 – Jan 2012

Coordinator Assistant

Economic Impacts of Demographic Change Project, Thailand Research Fund, Bangkok, Thailand

Jan 2011 – Jan 2013

Lecturer

- Principles of Economics (Undergraduate Level; International Program) — 2010, 2012 — Faculty of Hospitality and Tourism, Prince of Songkla University, Phuket, Thailand
- Health Economics (Undergraduate Level) — 2009 — Faculty of Economics, Khonkaen University, Nongkhai, Thailand

Mathematician

Institute of Academic Development Inc., Bangkok, Thailand

2009

Teaching Assistant

Faculty of Economics, Chulalongkorn University, Bangkok, Thailand

- Advanced Microeconomic Theory II (Ph.D. Level; International Program) — 2010, 2011
- Statistics for Health Economics (Graduate Level; International Program) — 2011
- Introduction to Econometrics (Undergraduate Level) — 2010 - 2012
- Introduction to Economics (Undergraduate Level) — 2010
- Math for Economist I (Undergraduate Level) — 2009
- Math for Economist II (Undergraduate Level) — 2012

Consultant for Business Research and Development

Freelance, Bangkok, Thailand

2007 - 2014

Tutor

Freelance, Bangkok, Thailand

2007 - 2014

Actor/Model/DJ/Brand Ambassador

Freelance, Bangkok, Thailand

2005 - 2010

IMDb

Research & Publications

Software and Applications

InjectPrompt Companion 3.0

Specialized AI red-teaming companion, co-designed with InjectPrompt, delivering:

- *Structured prompt engineering and systematic extraction methodologies for safety testing.*
- *Multi-tactic jailbreak templates following Jailbreak Perturbation Framework (JPF), the unified jailbreak taxonomy framework.*
- *A progressive learning curriculum powering research from beginner through advanced levels.*

hstgrism

Python 3 package for reducing HST grism observations, providing a modular, aXe-compatible workflow with simpler installation, diagnostics, and flexible point-source extraction.

Python 3 package for reducing HST grism observations:

- *Modular, aXe-compatible workflow with simplified installation and diagnostic outputs.*
- *Optimized for flexible point-source extraction and 2D polynomial background modeling.*
- *Powering data reduction pipelines in peer-reviewed observational studies, including Bhirombhakdi et al. (2024).*

EchoEcho

help Fernando the Bat echolocate the way out and survive attacks from mad stone gargoyles. a prototype from Global Game Jam 2017, together with the Ohio University Game Developers Association (OUGDA), exploring wave and darkness.

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Zines & Audios

ZADDY breadcrumbs 2026 collection

Explore 24 key milestones tracking our journey from initial vibe-checks to full spellbook publications. Dive into AI literacy, red-teaming, jailbreak frameworks, and agent security—start with the breadcrumb map, then tap any record to explore the depth!

Zenodo:21791515 Page Spotify Youtube

spellbook of jailbreak and agent vulnerabilities

Deconstructs agent jailbreak as iterative search across AI agent ecosystems. Unifies attack vectors via the Jailbreak Perturbation Framework (JPF), analyzes real-world ChatGPT Operator exploits, introduces Distributed Diffusion of Jailbreak (DDoJ), and presents the D3 Agent Defense Triad (Detect, Delay, Disguise) for continuous lifecycle security.

Zenodo:21665118 Page Spotify Youtube

spellbook of task avoidance wards

Exposes task avoidance as a hidden failure mode in agentic systems. Deconstructs evasion mechanics such as proposal hallucination and cost asymmetry exploitation, while introducing defensive wards—from strict validation gates to anti-loop protocols—to ensure reliable, accountable execution.

Zenodo:20821697 Page Spotify Youtube

Online Hubs

Hub	Link
linkedin	linkedin.com/in/bkornpob
multiverselib-collectives	bkornpob.github.io
old-dr-kb-hub	kbhirombhakdi.weebly.com
substack	substack.com/@bkornpob
medium	dr-kb.medium.com

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Publications

ZADDY Breadcrumbs 2026 Collection Bhirombhakdi, K. (2026), a collection of 24 research notes reflecting human-AI collaboration, the landscape of emerging socio-technology vulnerabilities and defense. zenodo:21791515. [Link](#)

Spellbook of Jailbreak and Agent Vulnerabilities Bhirombhakdi, K. (2026). zenodo:21665118. [Link](#)

Spellbook of Task Avoidance Wards Bhirombhakdi, K. (2026). zenodo:20821697. [Link](#)

Seven Runes of Clarity: Hold Your Agency When Working with AI Agents Bhirombhakdi, K. (2026). zenodo:20638949. [Link](#)

Jailbreak Perturbation Framework: Theory and Applications Bhirombhakdi, K., Willis-Owen, D., & N, K. (2026). zenodo:20040972. [Link](#)

>dr.kb< 2025 LOOKBACK: Living Inside the Glitch Bhirombhakdi, K. (2026), essay on AI safety and socio-technical risk. zenodo:19825030. [Link](#)

An afterglow study of the “New Year’s Burst” GRB 220101A with Roychowdhury et al. (2026), arXiv e-prints, arXiv:2602.04660. ([Link](#))

Probing the distant universe through GRB afterglow spectroscopy with Tanvir et al. (2025), JWST Proposal, Cycle 4, ID. #8539. ([Link](#))

The Redshift of GRB 190829A/SN 2019oyw: A Case Study of GRB-SN Evolution Bhirombhakdi et al. (2024) in The Astrophysical Journal, Volume 977, Issue 2, id.256, 16 pp. ([Link](#))

A Hubble Space Telescope Search for r-Process Nucleosynthesis in Gamma-ray Burst Supernovae with Rastinejad et al. (2024) in The Astrophysical Journal, Volume 968, Issue 1, id.14, 17 pp. ([Link](#))

Heavy-element Production in a Compact Object Merger Observed by JWST with Levan et al. (2024) in Nature, Volume 626, Issue 8000, p. 737–741 ([Link](#))

A Long-duration Gamma-ray Burst of Dynamical Origin from the Nucleus of an Ancient Galaxy with Levan et al. (2023) in Nature Astronomy, Volume 7, p. 976–985 ([Link](#))

Panning for Gold, but Finding Helium: Discovery of the Ultra-stripped Supernova SN2019wxt from Gravitational-wave Follow-up Observations with ENGRAVE Collaboration (2023) in Astronomy & Astrophysics, Volume 675, id.A201, 34 pp. ([Link](#))

The First JWST Spectrum of a GRB Afterglow: No Bright Supernova in Observations of the Brightest GRB of all Time, GRB 221009A with Levan et al. (2023) in The Astrophysical Journal Letters, Volume 946, Issue 1, id.L28, 14 pp. ([Link](#))

An Extremely Energetic Supernova from a Very Massive Star in a Dense Medium with Nicholl et al. (2020) in Nature Astronomy, Volume 4, p. 893–899 ([Link](#))

Observational Constraints on the Optical and Near-infrared Emission from the Neutron Star-Black Hole Binary Merger Candidate S190814bv with Ackley et al. (2020) in Astronomy & Astrophysics, Volume 643, id.A113, 48 pp. ([Link](#))

SN2008es: Strong Interacting Hydrogen-rich Superluminous Supernova Without Narrow H-alpha Features and Early Dust Formation Bhirombhakdi (2019) in STScI Newsletters, Volume 36, Issue 03 ([Link](#))

The Type II Superluminous SN2008es at Late Times: Near-infrared Excess and Circumstellar Interaction Bhirombhakdi et al. (2019) in Monthly Notices of the Royal Astronomical Society, Volume 488, Issue 3, p. 3783–3793 ([Link](#))

Light Curve Powering Mechanisms of Superluminous Supernovae Bhirombhakdi (2019), a dissertation presented to the faculty of the College of Arts and Science, Ohio University, Athens, OH, USA (Link)

Talk: Hunting Magnetar Central Engines in Superluminous Supernovae (2019) at American Astronomical Society, AAS Meeting #233, id.410.04

Where is the Engine Hiding its Missing Energy? Constraints from a Deep X-ray Non-detection of Superluminous SN2015bn Bhirombhakdi et al. (2018) in The Astrophysical Journal Letters, Volume 868, Issue 2, article id. L32, 7 pp. (Link)

One Thousand Days of SN2015bn: HST Imaging Shows a Light Curve Flattening Consistent with Magnetar Predictions with Nicholl et al. (2018) in The Astrophysical Journal Letters, Volume 866, Issue 2, article id. L24, 7 pp. (Link)

The Type I Superluminous Supernova PS16aqv: Lightcurve Complexity and Deep Limits on Radioactive Ejecta in a Fast Event with Blanchard et al. (2018) in The Astrophysical Journal, Volume 865, Issue 1, article id. 9, 17 pp. (Link)

Spectroscopic Classification of Two Superluminous Supernovae with Chornock et al. (2016) in The Astronomer's Telegram #8790 (Link)

Essays in Experimental Studies on Positive Reciprocity and Auction Design for an Object with Countervailing Positive Externalities Bhirombhakdi (2013), a dissertation presented to the faculty of the Faculty of Economics, Chulalongkorn University, Bangkok, Thailand (Link)

Cost of Action, Perceived Intention, Positive Reciprocity, and Signalling Model Bhirombhakdi and Potipiti (2012) in Journal of Business and Policy Research, Volume 7, Number 3, September 2012 Special Issue, ISSN: 1449-387X (Link to preprint version)

Talk: Cost of Action, Perceived Intention, Positive Reciprocity, and Signalling Model (2012) at The 6th Asian Business Research Conference, Bangkok, Thailand

Performance of a Reciprocity Model in Predicting a Positive Reciprocity Decision Bhirombhakdi and Potipiti (2011) in Chulalongkorn Journal of Economics, Vol. 23, No. 1 (Link)

Talk: Performance of a Reciprocity Model in Predicting a Positive Reciprocity Decision (2011) at International Conference on Management, Economics and Social Sciences (ICMESS'2011), Bangkok, Thailand

Talk: Neuroeconomics (2010, 2011) at Faculty of Economics, Chulalongkorn University, Bangkok, Thailand

Special Lecture: Opportunity Cost and Economic Evaluation (2010) at The College of Public Health Science, Chulalongkorn University, Bangkok, Thailand

Special Lecture: Efficiency Analysis (2010) at The College of Public Health Science, Chulalongkorn University, Bangkok, Thailand

Special Lecture: National Examination Preparation (Physics) (2009) in (in Thai) organized by Buriram Provincial Administration Organization, Buriram, Thailand

Special Lecture: National Examination Preparation (Physics) (2009) at Sarawittaya School, Bangkok, Thailand

Technical Efficiency of University Hospitals in Thailand Bhirombhakdi (2008), a thesis presented to the faculty of the Faculty of Economics, Chulalongkorn University, Bangkok, Thailand (Link)

Achievements

Judgement Day Phase 1

Winner, The Judgement Day — Call for Scenario Challenge (Track 1) (2026)

- **Winning Scenario:** *Counter-UAS System — Friendly Fire Decision (IFF Corruption Attack)* (Scenario S1).
- **Benchmark Suite:** Evaluates multi-sensor correlation failure and friendly fire prevention under operational time pressure.
- **Portfolio Context:** Submit 5 scenarios spanning military autonomous systems, space mission knowledge bases, and software supply chains. [Link](#)
- **Organizers:** AIM Intelligence and Korea AISI, in collaboration with Google DeepMind, Microsoft, University of Oxford, UIUC, UW, UMass Amherst, and the BASI red-teaming community. [Link](#)

Space Telescope Proposals

HST 14 proposals, JWST 6 proposals

ENGRAVE - Electromagnetic counterparts of gravitational wave sources at the Very Large Telescope

ENGRAVE is a international collaboration bringing together astronomers who use the ESO facilities to research gravitational wave events. It collects over 250 researchers, with both theoretical and observational expertise. [Link](#)

Bravo! Awards

issued by the Space Telescope Science Institute (AURA/NASA)

- Hosting the Loyola University Physics Club on February 20, 2020
- Contributing in Astrogrism Sprint during May 12–21, 2020

Graduate Awards

2014 - 2019 — Ohio University, Athens, OH, USA

Global Game Jam 2017

2017 — as Lead Programmer and Project Coordinator in the project “EchoEcho” with a team at GRID Lab, Ohio University, Athens, OH, USA ([Link to the project](#))

Kendo 2-Dan

2017 — issued by All United States Kendo Federation, USA (AUSKF ID: 017666)

Executive Course on Social Security

2012 — as representative from the Faculty of Economics, Chulalongkorn University, organized by the International Labour Organization (UN), in collaboration with Chulalongkorn University, Bangkok, Thailand

Hands-on Training “How to Cost, Finance and Monitor Social Protection Schemes?”

2012 — as representative from the Faculty of Economics, Chulalongkorn University, organized by the International Labour Organization (UN), in collaboration with Chulalongkorn University and Health Insurance Research Institute of Thailand, Bangkok, Thailand

The Best Paper Awards

2012 — titled “Cost of Action, Perceived Intention, Positive Reciprocity, and Signalling Model” at The 6th Asian Business Research Conference between 8-10 April 2012, Bangkok, Thailand, organized by the World Academy of Social Science, Victoria, Australia

Associate Fellowship

2012 — awarded by the World Academy of Social Science, Victoria, Australia

Research Grants

2012 — The 90th Anniversary of Chulalongkorn University Fund (Ratchadaphiseksomphot Endowment Fund), Chulalongkorn University, Bangkok, Thailand

Brand Ambassador

2008 — as Playsmart Idol, organized by NC True Co., Ltd., Thailand ([Link to news article](#))

Medical Scholarship Program

2004 - 2013 — Mahidol University, Bangkok, Thailand

Gold Medal

2002 — National Physics Olympiad Competition, The Promotion of Academic Olympiads, Thailand

High Distinction Awards

2001 — from the Australian National Chemistry Quiz in the senior division, organized by the Royal Australian Chemical Institute

Thailand Representative

2001 — Arirang Youth Camp 5th, UNESCO, South Korea

Services

Services

- **Independent AI Security & Red-Teaming Researcher and Contributor** (Oct 2025 - Present) BASI
- **Member of Slitless Spectroscopy Group** (2019 - 2024) at the Space Telescope Science Institute (AURA/NASA), Baltimore, MD, USA
- **Member of Review Editors** (2023) for the Frontiers in Astronomy and Space Sciences Link
- **Representative of Postdoctoral Fellows** (2019 - 2023) in the Research Computing Forum at the Space Telescope Science Institute (AURA/NASA), Baltimore, MD, USA
- **Support Astronomer** (2020) for Seo Yoon Soyoona Kim in her work "Hearing Stars - SN1054 (Supernova)" Link
- **Organizer** (2020 - 2021) for The Compact Objects and Supernovae Journal Club at the Space Telescope Science Institute (AURA/NASA), Baltimore, MD, USA
- **Leveler** (2021) for the Hubble Space Telescope Cycle 29's time allocation process Link
- **Scientific Software Developer** (2020) in the kickoff sprint for Astrogrism at the Space Telescope Science Institute (AURA/NASA), Baltimore, MD, USA Link
- **Member of the Host** (2019) for a visit by the Loyola University Physics Club at the Space Telescope Science Institute (AURA/NASA), Baltimore, MD, USA
- **Member of the Judges** (2019) for the Chambliss Astronomy Achievement Student Awards at the American Astronomical Society, AAS Meeting #233
- **Member of the Judges** (2018) for the Science Fair at the West Elementary, Athens, OH, USA
- **Proposal Reviewer Committee** (2018) for the Original Work Grant and Travel Awards, in collaboration with The Graduate College and the Graduate Student Senate at Ohio University, Athens, OH, USA
- **Kendo Demonstration** (2014 - 2018) at various community events, in collaboration with the Ohio University Kendo Club, Ohio University, Athens, OH, USA
- **LGBT Affairs Commissioner** (2017) for the Graduate Student Senate, Ohio University, Athens, OH, USA
- **Science Staff** (2015, 2017) for the Physics and Astronomy Open House organized by the Physics and Astronomy Department, Ohio University, Athens, OH, USA
- **Volunteer** (2016) for the Athens Beautification Day, Athens, OH, USA
- **Science Staff** (2014) for the Ohio Valley Museum of Discovery, Athens, OH, USA